

PART 4: SEQUENTIAL LOGIC

SYNCHRONOUS COUNTERS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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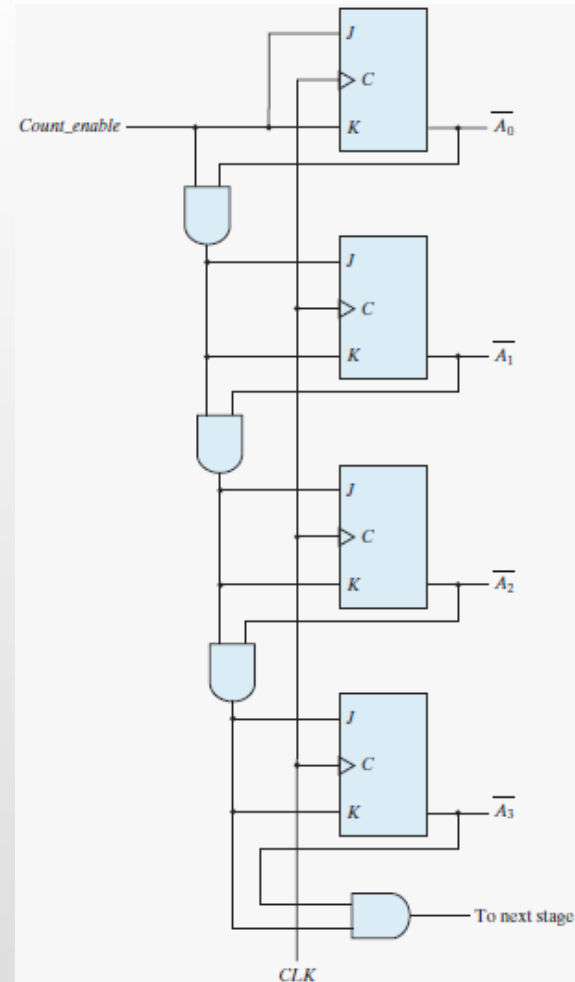
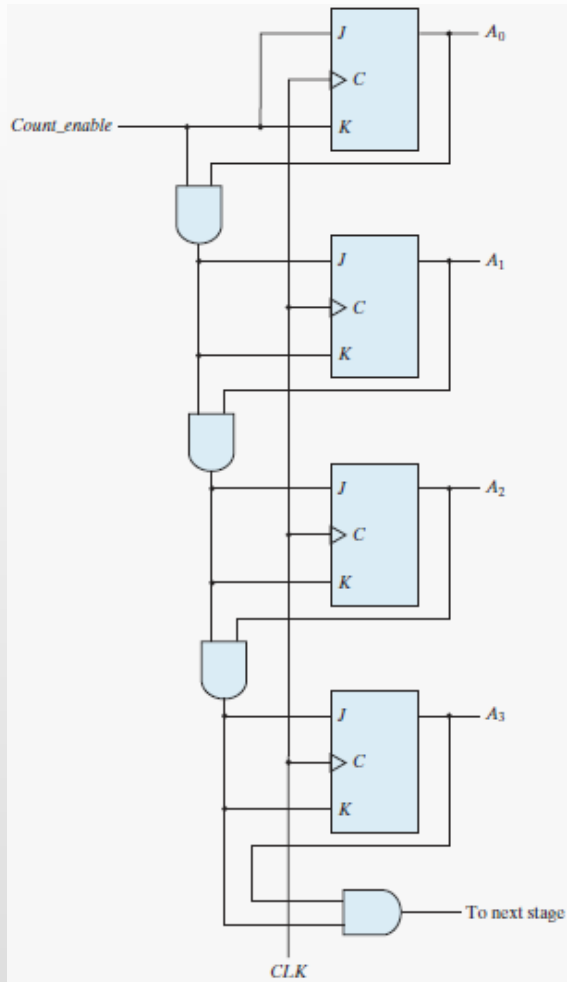
SYNCHRONOUS COUNTERS

- Synchronous counters are different from ripple counters in that clock pulses are applied to the inputs of all flip-flops.
- A common clock triggers all flip-flops simultaneously, rather than one at a time in succession as in a ripple counter.
- The decision whether a flip-flop is to be complemented is determined from the values of the data inputs, such as T or J and K at the time of the clock edge.

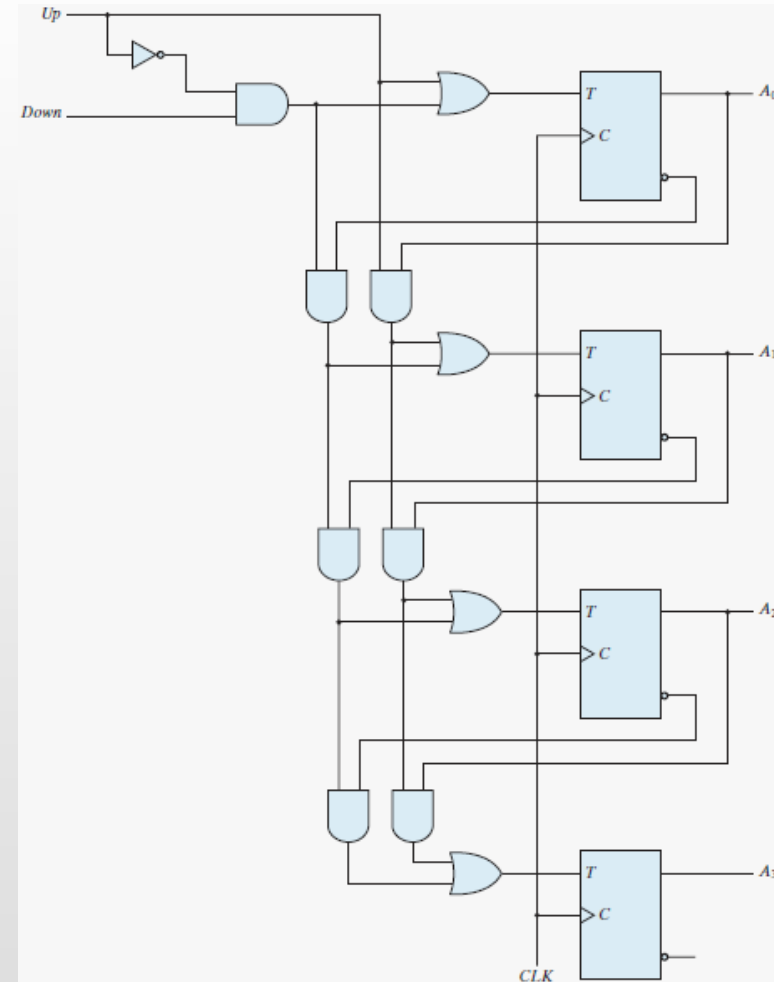
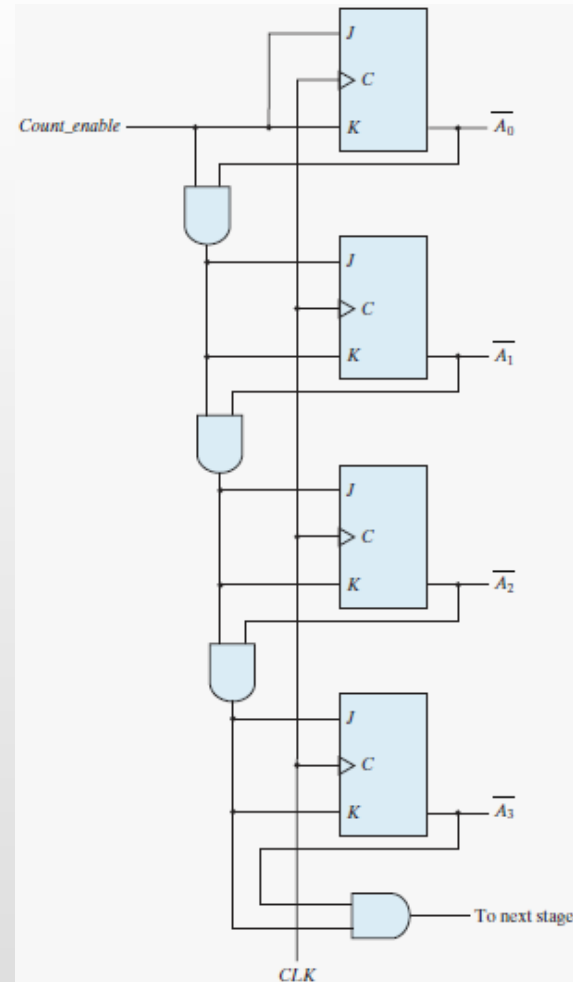
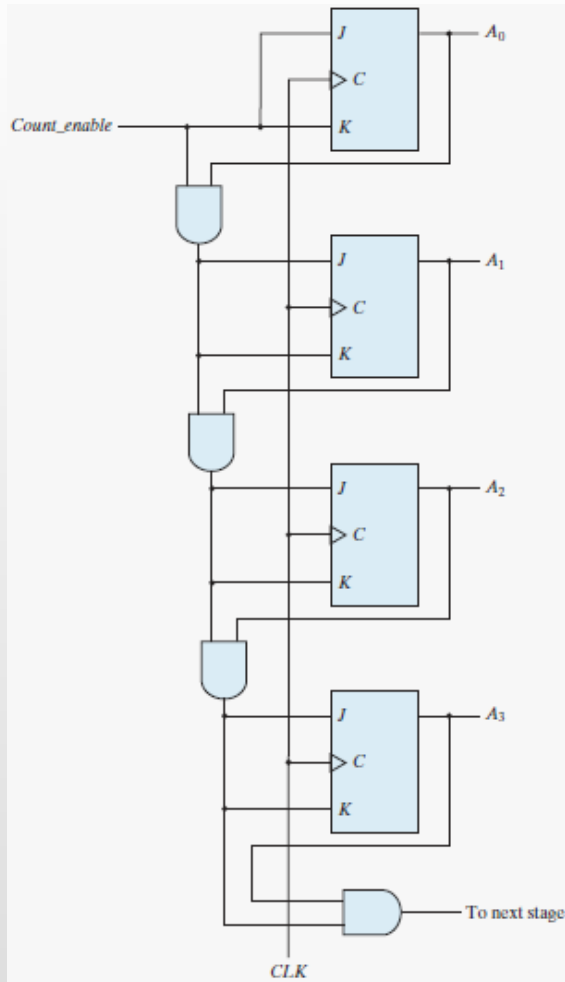
BINARY COUNTER



- The design of a synchronous binary counter is so simple that there is no need to go through a sequential logic design process.
- In a synchronous binary counter, the flip-flop in the least significant position is complemented with every pulse.
- Synchronous binary counters have a regular pattern and can be constructed with complementing flip-flops and gates.

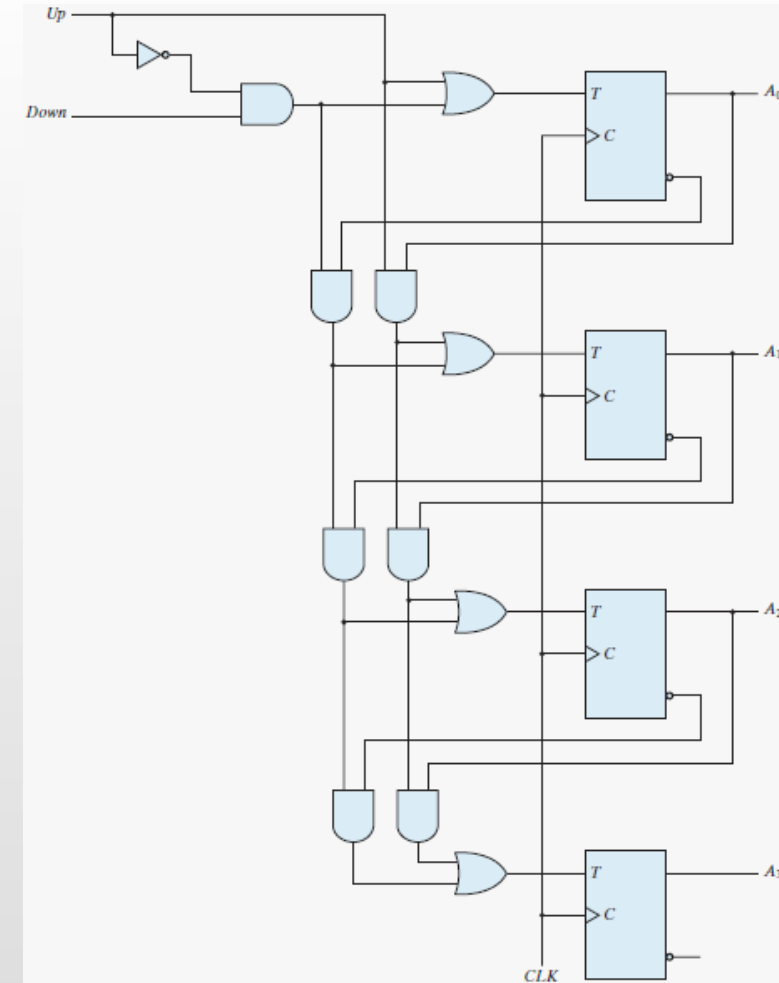


BINARY SYNCHRONOUS DOWN COUNTER



BINARY SYNCHRONOUS UP-DOWN COUNTER

- When the up input is 1, the circuit counts up.
When the down input is 1 and the up input is 0, the circuit counts down.
- When the up and down inputs are both 0, the circuit does not change state and remains in the same count.
- What happens when the up and down inputs are both 1?
 - The circuit counts up.
 - The up input has priority over the down input.

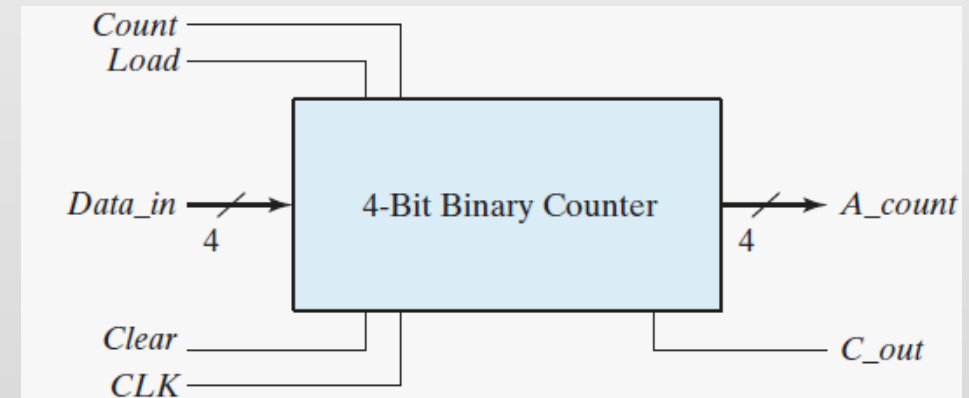


BINARY SYNCHRONOUS UP-DOWN COUNTER

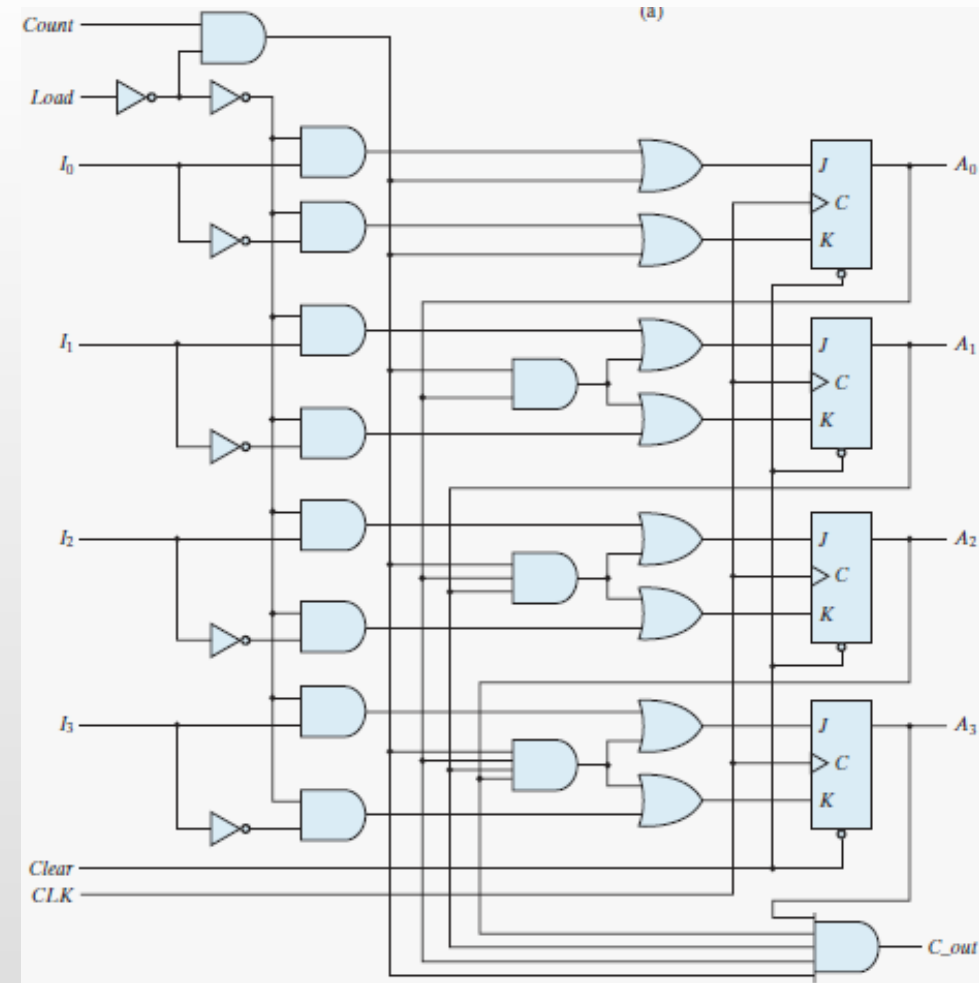
BINARY COUNTER WITH PARALLEL LOAD

- Counters employed in digital systems quite often require a parallel-load capability for transferring an initial binary number into the counter prior to the count operation.
- A counter with a parallel load can be used to generate any desired count sequence.

Clear	CLK	Load	Count	Function
0	X	X	X	Clear to 0
1	↑	1	X	Load inputs
1	↑	0	1	Count next binary state
1	↑	0	0	No change



Clear	CLK	Load	Count	Function
0	X	X	X	Clear to 0
1	↑	1	X	Load inputs
1	↑	0	1	Count next binary state
1	↑	0	0	No change



4-BIT BINARY COUNTER WITH PARALLEL LOAD

SYNCHRONOUS BCD COUNTER

- $y = 1$ when the present state is 1001 to enable the count of the next-higher significant decade. The same pulse switches the present decade from 9 to 0.

State Table for BCD Counter

Present State				Next State				Output	Flip-Flop Inputs			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	y	TQ_3	TQ_2	TQ_1	TQ_0
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1

SYNCHRONOUS BCD COUNTER

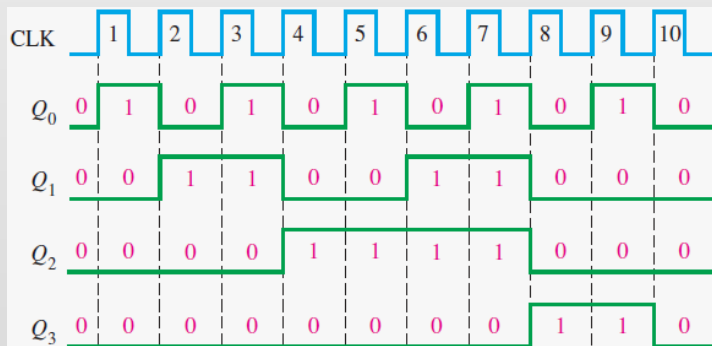
State Table for BCD Counter

Present State				Next State				Output	Flip-Flop Inputs			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	y	TQ_3	TQ_2	TQ_1	TQ_0
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
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1	0	0	1	0	0	0	0	1	1	0	0	1

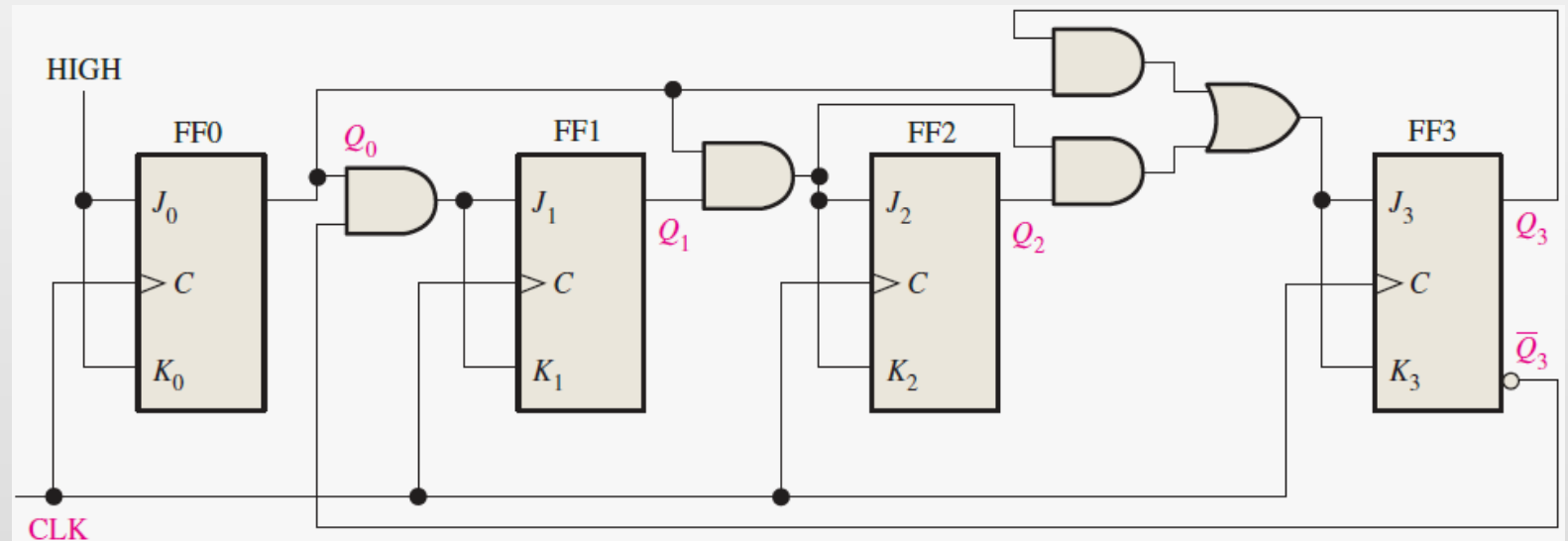
- $J_0 = K_0 = 1$
- $J_1 = K_1 = Q_0 \overline{Q_3}$
- $J_2 = K_2 = Q_0 Q_1$
- $J_3 = K_3$
 $= Q_0 Q_1 Q_2 + Q_0 Q_3$

4-BIT SYNCHRONOUS DECADE COUNTER

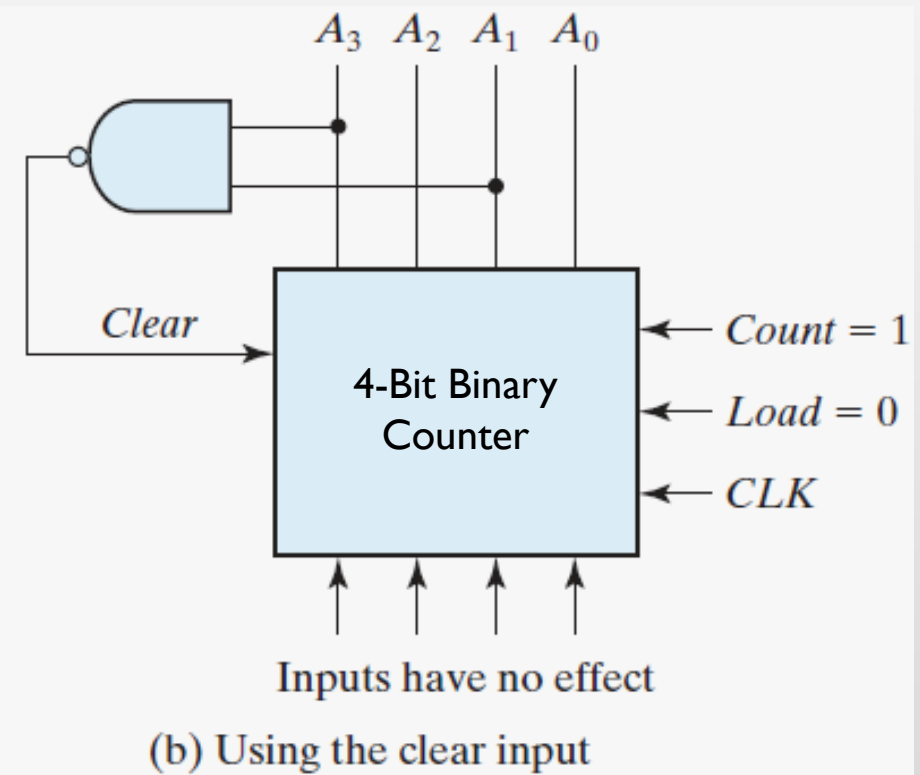
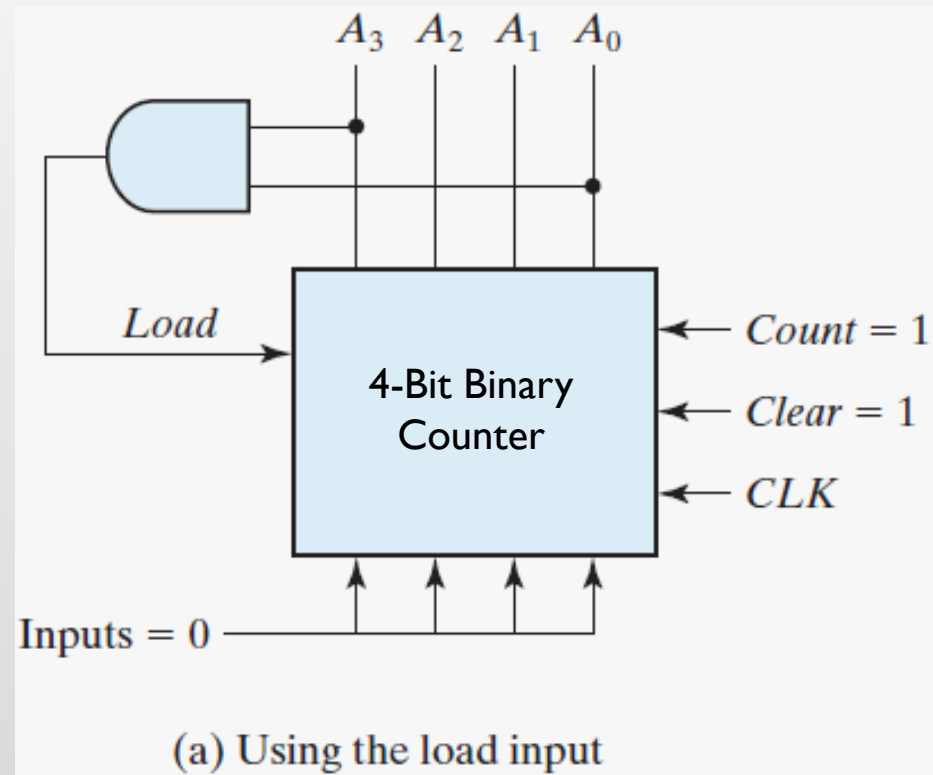
- $J_0 = K_0 = 1$
- $J_1 = K_1 = Q_0 \overline{Q_3}$
- $J_2 = K_2 = Q_0 Q_1$
- $J_3 = K_3 = Q_0 Q_1 Q_2 + Q_0 Q_3$



Timing diagram for the BCD decade counter (Q_0 is the LSB).

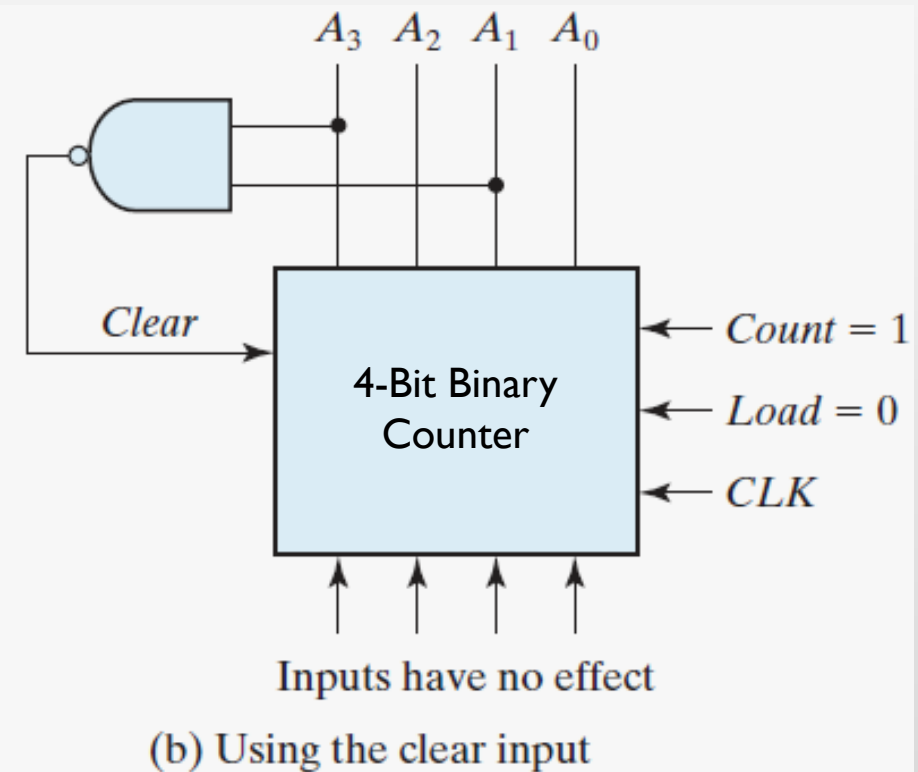


BCD COUNTER USING A COUNTER WITH PARALLEL LOAD

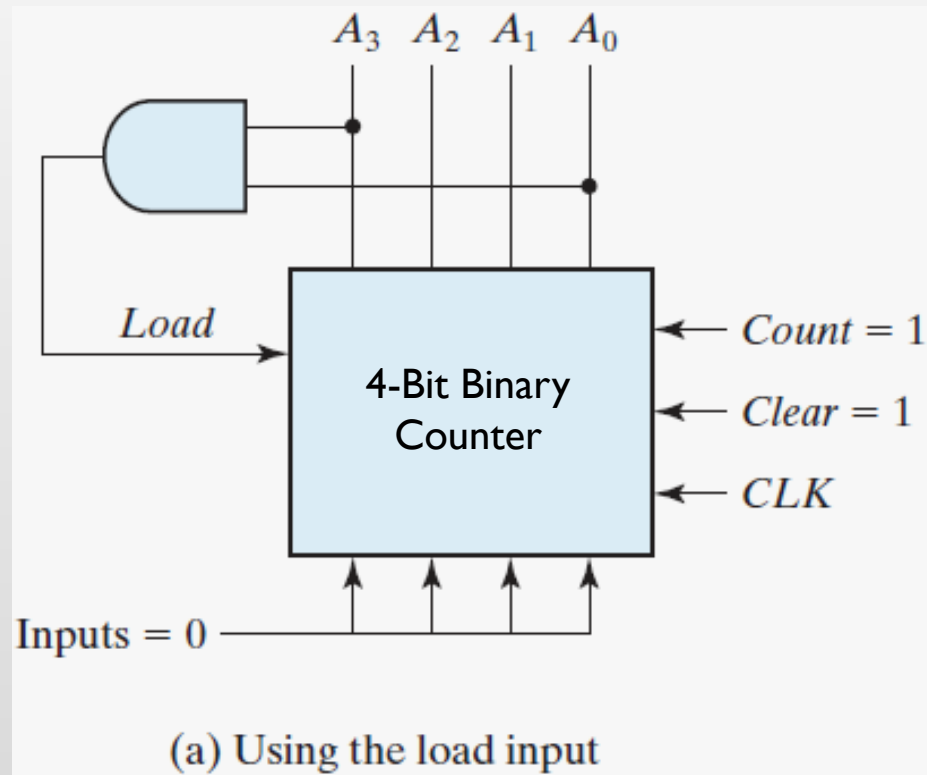


BCD COUNTER USING A COUNTER WITH PARALLEL LOAD

- The NAND gate detects the count of 1010, but as soon as this count occurs, the register is cleared.
- A momentary spike occurs in output A_0 as the count goes from 1001 to 1010 and immediately to 0000. For that reason, this configuration is not recommended.
- If the counter has a synchronous clear input, it is possible to clear the counter with the clock after an occurrence of the 1001 count.



BCD COUNTER USING A COUNTER WITH PARALLEL LOAD



- As long as the output of the AND gate is 0, each positive-edge clock increments the counter by 1.
- When the output reaches the count of 1001, both A_0 and A_3 become 1, making the output of the AND gate equal to 1.
- This condition activates the *Load* input. On the next clock edge an all-0's value is loaded into the register following the count of 1001. Thus, the circuit goes through the count from 0000 through 1001 and back to 0000, as is required in a BCD counter.